

Practical Number : 7

1. Title: Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

2. Software/Tools Required: Visual Studio Code (VSCode), JavaScript.

3. Theory:

DOM (Document Object Model) traversal and manipulation allow JavaScript to dynamically select, structure, and modify webpage content

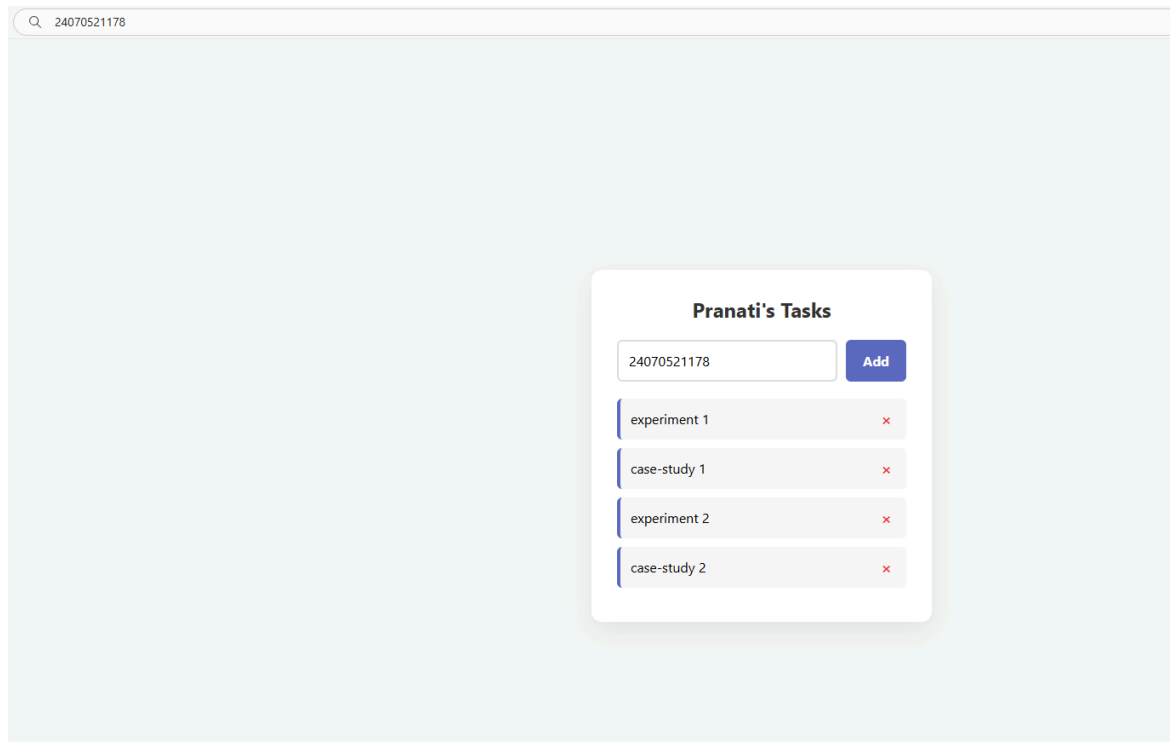
without requiring a full page refresh.

DOM Traversal: Involves navigating through parent, child, and sibling nodes using methods like `getElementById()`, `createElement()`, and `appendChild()`.

Dynamic DOM Updates: Enables creating elements on the fly (`createElement('li')`, `(classList.toggle('completed'))`, and removing elements from the node tree (`li.remove()`).

Event Handling: Listens for user interactions (click, keypress) using `addEventListener()` to trigger functions that mutate the structure and appearance of the page in real time

4. Output:



5. Result/Conclusion:

The To-Do List application demonstrates how JavaScript interacts with the DOM tree to construct interactive web interfaces. By combining event listeners with DOM traversal, creation, and deletion methods, the application handles user inputs, dynamically updates task statuses, and modifies the DOM structure seamlessly.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study : 7

1. Title: To Demonstrate Event Handling in JavaScript Using Form Elements and Input Validation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Event handling in JavaScript is used to respond to user actions on a webpage. In this experiment, HTML is used to create a student registration form, CSS is used for styling, and JavaScript is used to access and manipulate form elements. Different events such as focus, change, and submit are implemented. JavaScript validates user input, checks for empty fields, and verifies the email format before displaying the result.

4. Output:

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'JavaScript Event Handling'. The address bar shows the URL '127.0.0.1:5500/index.html'. The main content area displays a 'Student Registration Form' with the following fields:

- Name:** A text input field with a yellow highlight.
- Email:** A text input field with a yellow highlight.
- Course:** A dropdown menu with 'Computer Science' selected.
- Submit:** A green button.

Below the form, a green message states: 'Form submitted successfully!'.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Conclusion : In this experiment, we successfully implemented event handling in JavaScript using a student registration form. We learned how to access form elements, handle focus, change, and submit events, and validate user inputs. This experiment helped us understand how JavaScript makes webpages interactive and user-friendly.